

Editorial for February 2012 Issue

I. SPECIAL ISSUE PROPOSALS ARE ALWAYS WELCOME

PROPOSALS for special issues on topics of current interest in engineering, technology, and innovation management are always welcome. Please send a brief description of the concept for the issue to me (sabherwal@ieee.org). I will circulate it to Department Editors, and if the initial response is favorable, I will request a specific plan and more detailed information to be used in the final decision about proceeding with the special issue.

II. ONLINE SUBMISSION AND REVIEW SYSTEM

The journal office currently experiences many returned e-mails via the journal's online system. Since e-mails are used to communicate with corresponding authors, coauthors, and reviewers during the review process, it is very important for users to keep updating their accounts in the system. Please login to <http://mc.manuscriptcentral.com/tem-ieee> to update your contact information when you change your e-mail addresses. New users are also welcome to create their own accounts in the system.

Also, please be sure that the SPAM filters on your own servers are set to receive mail from Manuscript Central. The journal's online server should be whitelisted to mark it as a valid e-mail sender: 170.107.181.135.

III. ABOUT THIS ISSUE

This issue of the IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT includes 10 research articles, 1 research note, and 1 focus on practice paper. Each of these papers is briefly described below.

"Networking Capability and New Product Development" (by Mu and Di Benedetto) Network relationships can allow firms to create value that no single firm could have created alone. The results of this study indicate that networking capability can profoundly impact new product development performance. Firms can influence the interaction patterns they face rather than take them as given. Firms can play the role of the strategic center in interfirm networks to manage their web of partners over time to sustain their innovativeness. Innovation is no longer just a matter of hiring the most talented and creative people or creating the right internal environment for it. In addition, innovation needs firms to understand how to design networks and how to tap into their potential. Firms need to build networking capability to get access into complementary resources that may be controlled by other actors in the network. This ability has become even more important in the last decade as firms have become leaner and their ability to leverage their own presence in various industries

or markets has been reduced. Firms must take strategic action to form and maintain a network and to extract value from it. Firms can, and should, use external, as well as internal ideas and resources to advance technology and product innovation.

"Lean Software Management: BBC Worldwide Case Study" (by Middleton and Joyce) This lean software case study is useful to: software developers who wish to improve their discipline and productivity; corporations who use lean to manage their manufacturing operations; and managers obliged to use the Capability Maturity Model Integrated (CMMI). This paper offers three main lessons: 1) The data suggests that lean quickly enables an intrinsically lower risk and more productive approach to developing software which will increase profits. 2) Using lean for both manufacturing and software development provides a common approach and language, therefore, simplifying management of a corporation. 3) The description of statistical process control in lean software could allow managers to quantify their software development process that may enable them to achieve CMMI level 4. This would allow them to work on Department of Defense and other contracts requiring CMMI certification.

"Personalization or Codification? A Marketing Perspective to Optimize Knowledge Reuse Efficiency" (by Chai and Nebus) This paper proposes a metric, Knowledge Reuse Efficiency (KRE), that can be used by practitioners and academics to gauge the effectiveness of knowledge management programs by taking an organization-wide view of knowledge reuse. The paper shows that the choice of a codification versus personalization strategy, two common knowledge management strategies, should be guided by the proposed KRE metric on a per stage basis. For example, an organization's knowledge management strategy to efficiently support an awareness or cursory understanding of knowledge among many consumers in the early stages of knowledge sharing may not also be the most efficient strategy when seeking to facilitate a deep understanding of that knowledge among comparatively few consumers in the later stages.

"Managing Risks in Just-in-Sequence Supply Networks: Exploratory Evidence from Automakers" (by Wagner and Silveira-Camargos) Given the popularity of just-in-sequence (JIS) delivery in industrial practice and the increasing vulnerability of JIS supply networks (SNs) to disruptions, managers need to understand how JIS delivery influences the risk exposure of their firms' SNs, and what they can do to mitigate these risks. This study helps managers to better understand and categorize current practice of JIS logistics strategies. It analyzes demand-side, internal, and supply-side risk sources in JIS delivery systems, and shows managers where the greatest risks are that they should be aware of. Finally, specific recommendations are provided that help to manage and mitigate these potential risks in order to avoid costly supply chain disruptions.

“Intellectual Human Capital and the Emergence of Biotechnology: Trends and Patterns, 1974–2006” (by Hess and Rothaermel) This paper presents an analysis of the trends and patterns documenting the role of intellectual human capital in the emergence of knowledge within both new biotech ventures and incumbent pharmaceutical firms. The paper provides insights into the similarities and differences between the activity, productivity, and movement of star and nonstar scientists across both biotech and pharma firms, and over time. Such an analysis provides unique insights into the importance of considering the differences within intellectual human capital and the roles that scientists played (and continue to play) in spreading the knowledge underlying biotechnology, not only throughout firms, but also throughout the scientific community as a whole.

“Relationships Between Leadership and Success in Different Types of Project Complexities” (by Muller, Gerald, and Turner) This article contributes to practice in three ways. Firstly, it allows engineering managers to advance their practices for assigning suitable (best-fit) project managers to projects, depending on project manager leadership competences and project complexity. The paper contributes the next level of detail to the current debate on project manager to project fit. The results from the present study allow human resource managers to recruit or develop project managers to link personality and project type appropriately in order to improve project performance and ultimately organizational performance. Secondly, the paper provides a statistically tested model to assess the context of projects in terms of project complexity. Engineering managers can apply the model to identify the idiosyncrasies of the complexities of their particular projects and subsequently chose best-fit leaders for their projects. Thirdly, the paper examines the differences between engineering projects and information technology (IT) and organizational change projects in respect of complexity measurements. Commonalities across project types indicate that the types of complexity have the potential to act as common language across industries and project types, or at least within the types and sectors analyzed, thus, facilitating cross-industry knowledge sharing and learning.

“Time Separation, Coordination, and Performance in Technical Teams” (by Espinosa, Cummings, and Pickering) This study was motivated by one company’s managerial interest in working effectively and productively across time zones. Work is becoming more and more global over time with a substantial amount of work being conducted across geographic locations. The growth of globally delivered IT services is expected to soon surpass the growth of IT services as a whole. This trend is driven partly by potential cost reductions and larger availability of technical talent in various parts of the globe. But these benefits can be quickly offset by high coordination costs when projects face substantial coordination problems. While global organizations have developed some expertise working across distance through technology, having to work across multiple time zones that span great time zone differences continues to be a big problem for management because the workflow needs to be more carefully coordinated to make effective use of overlapping and nonoverlapping work times. This study helps management understand how time separation across time zones impacts coordination

problems, how these problems influence team performance in turn, and how these coordination problems can be mitigated with effective task organization practices.

“Managing the Concurrent Execution of Dependent Product Development Stages” (by Lin, Qian, and Cui) Concurrent product development, the practice of executing dependent product development stages simultaneously, has become the common mode of new product development because of the increasing importance of time to market. However, such simultaneous execution of dependent stages may substantially increase the total amount of rework. Consequently, an analytical investigation of the tradeoffs involved in concurrent process is needed. This paper indicates that initial development is prior to rework when learning effect is taken into account, which can be used by project managers to guide resource allocation among development activities. Moreover, the paper presents analytical models to help management determine the optimal concurrency of development stages.

“Interaction of Individual and Social Antecedents of Learning Effectiveness: A Study in the IT Research Context” (by Kankanhalli, Pee, Tan and Chhatwal) In knowledge-intensive environments, such as research units in IT firms or in universities, individual learning is key to organizational performance. However, learning can be arduous and its effectiveness is likely to depend on a combination of individual and social factors. With the current lack of understanding of their interactions, organizations are hampered in their attempts to promote learning. Therefore, the objective of this study is to explore the individual and social antecedents and their interaction effects on individual learning. The findings of this study identify the antecedent factors that work on their own or in specific combinations to affect the learning effectiveness of senior undergraduates working on IT-related research projects. While the results inform research and practice, it may be worthwhile to further validate them in organizational R&D settings.

“Stochastic Entry of Competitors and Marketing Decisions” (by Huang and Ho) This study assumes that the decision of keeping or stopping selling a product is essentially affected by the entry process of rival competitors within the planned product life. The proposed Bayesian analysis approach can provide managers with an informative decision process in formulating an appropriate strategy to maximize the product profits in a competitive market from the viewpoint of an incumbent firm.

“Assessing Organizational Capabilities: Reviewing and Guiding the Development of Maturity Grids” (by Maier, Moultrie, and Clarkson) This research note reviews and guides the development of maturity grid assessment methods to improve organizational capabilities and conduct change initiatives. This paper makes four contributions to practice. Firstly, a roadmap describing four phases—planning, development, evaluation, and maintenance—decision points and options for each phase to help practitioners evaluate existing maturity grids and guide them in the development of new methods tailored to their own needs and specific situation. Secondly, to provide a practical example of the roadmap’s utility in practice, the paper describes the development and application of a maturity grid to assess communication at the interface of two engineering departments

in the automotive industry. Thirdly, the research note presents a cross-domain literature review of existing maturity grids from a number of domains, including knowledge management, communication, product development, innovation and from a number of disciplines, such as engineering, healthcare, and management. Lastly, to ease navigation in the complex field of performance assessment, the research note clarifies terms associated with maturity in organizational practice, e.g., process maturity, organizational maturity, process capability, project maturity, and maturity of organizational capabilities.

“Empirical Investigation of EachNet: The eBay Model of C2C Online Auction in China” (by Du, Yu, Fang, and Wang)
Based on the results of this study, online auction sellers are advised in this focus-on-practice article to set proper auction

parameters (e.g., start bidding price, hidden reserve price, buy-it-now option), and monitor the bidding process in order to achieve auction success, effectiveness, and satisfaction. This study also suggests that product-independent performance measures should be used to assess the overall online auction performance in a given marketplace, instead of merely relying on the final auction price, which may vary significantly among different product categories.

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